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Course –ECA5601- 5g/6g Communication

Experiment: 10

5G Network Slicing Environment Using MATLAB

Aim

To analyze the performance of a 5G Network Slicing environment by studying the number of network slices, slice utilization, and end-to-end latency using MATLAB.

Objectives

Objective – 1: Analyze the Number of Network Slices

Analyze the creation and allocation of multiple network slices in a 5G network using MATLAB. Observe how different slices can be configured to support diverse service requirements and applications.

Objective – 2: Compare Slice Utilization (%)

Compare the utilization of different network slices using MATLAB. Analyze how efficiently network resources are allocated and utilized across multiple slices.

Objective – 3: Analyze End-to-End Latency (ms)

Analyze the end-to-end latency of different network slices using MATLAB. Compare the latency values to understand the Quality of Service (QoS) requirements of various 5G applications.

Procedure

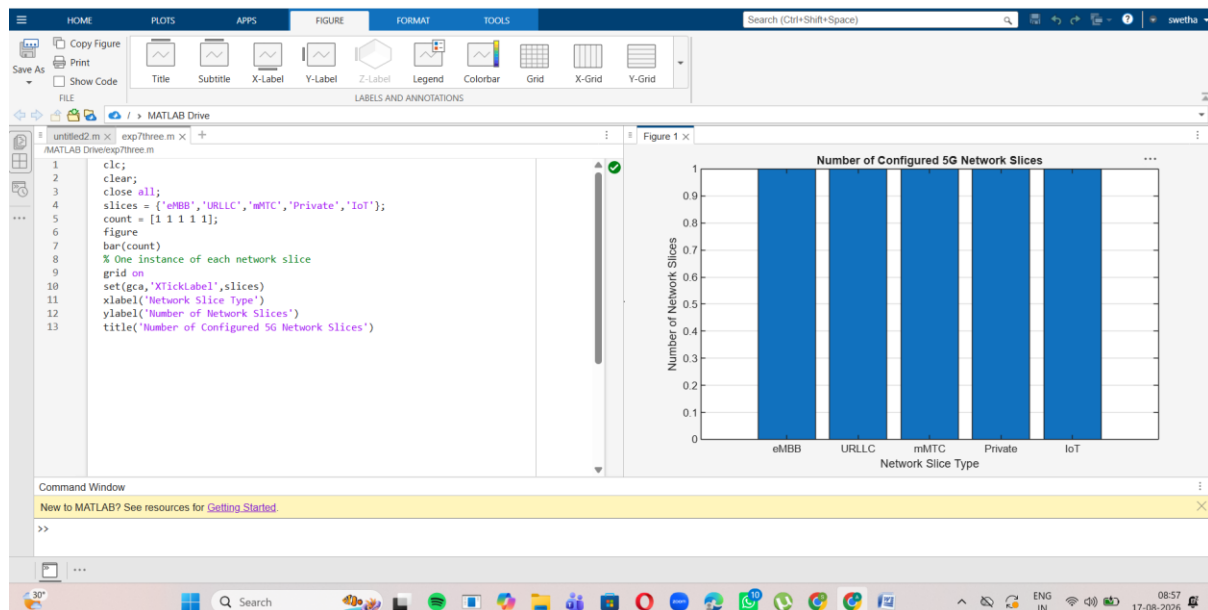
1. Define the network parameters such as available bandwidth, number of users, traffic load, and different types of network slices.
2. Create separate slices in MATLAB for applications such as enhanced mobile broadband, ultra-reliable low-latency communication, and massive IoT.
3. Allocate network resources such as bandwidth and processing capacity to each slice based on its requirements.
4. Simulate different traffic conditions and observe how the resources are distributed among the network slices.
5. Plot and analyze the performance of each slice based on parameters such as resource utilization, throughput, and latency.

objective 1 :

Program

```
clc;
clear;
close all;
slices = {'eMBB','URLLC','mMTC','Private','IoT'};
count = [1 1 1 1 1];
figure
bar(count)
% One instance of each network slice
grid on
set(gca,'XTickLabel',slices)
xlabel('Network Slice Type')
ylabel('Number of Network Slices')
title('Number of Configured 5G Network Slices')
```

output



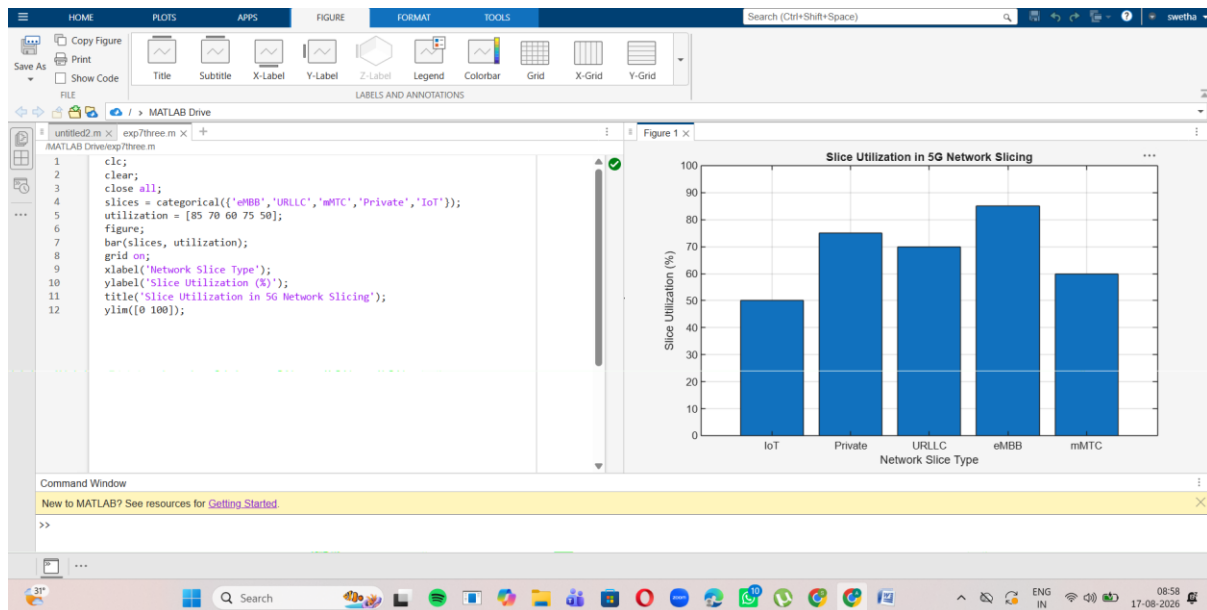
Objective 2: code and output:

```
clc;
clear;
close all;
slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});
utilization = [85 70 60 75 50];
figure;
bar(slices, utilization);
grid on;
```

```

xlabel('Network Slice Type');
ylabel('Slice Utilization (%)');
title('Slice Utilization in 5G Network Slicing');
ylim([0 100]);

```

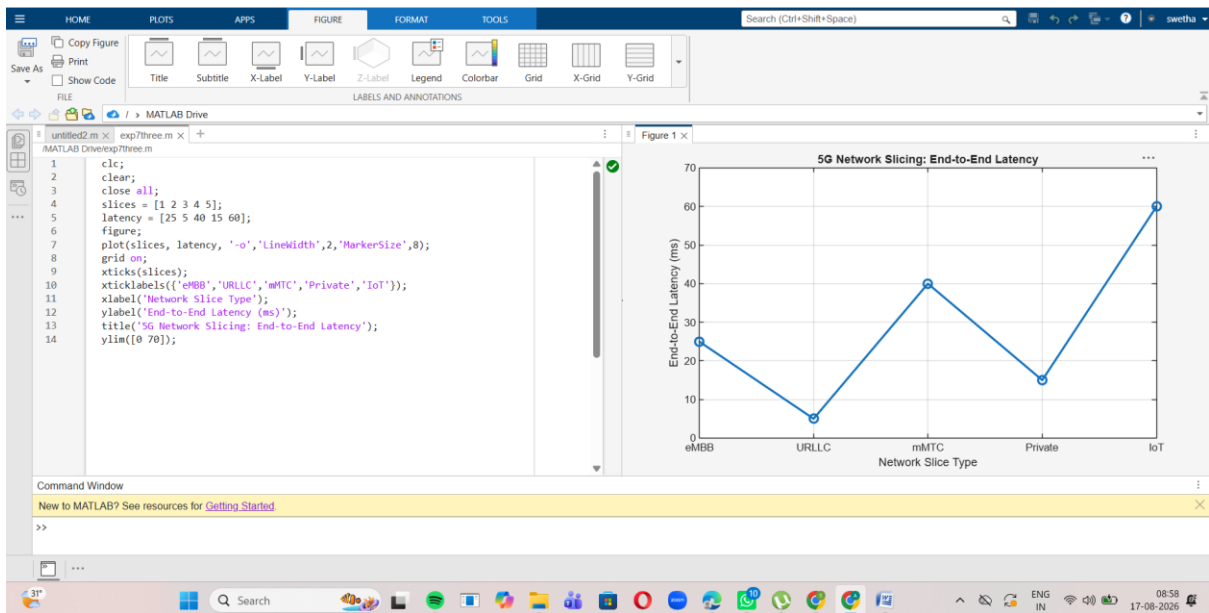


Objective 3:code and output:

```

clc;
clear;
close all;
slices = [1 2 3 4 5];
latency = [25 5 40 15 60];
figure;
plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8);
grid on;
xticks(slices);
xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});
xlabel('Network Slice Type');
ylabel('End-to-End Latency (ms)');
title('5G Network Slicing: End-to-End Latency');
ylim([0 70]);

```



Result:

The MATLAB analysis successfully demonstrated the characteristics of a 5G Network Slicing environment. The results showed that multiple network slices can coexist while exhibiting different utilization levels and latency characteristics based on their intended services.